

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Summer Examination – 2025****Course: B. Tech****Branch : BTCVC604_Y18****Semester : VI****Subject Code & Name: BTCVC604_Y18/ Project Management****Max Marks: 60****Date:08/07/ 25****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	To draw a CPM network effectively, which of the following must be known? a. Only activity durations b. Only sequence of activities c. Activity durations and their dependencies d. Start and end time of project		1
2	Which type of float indicates the delay that will not affect the succeeding activity? a. Free Float b. Total Float c. Independent Float d. Critical Float		1
3	A time-scaled network diagram is essential during: a. Scheduling b. Network development c. Project updating d. Budgeting		1
4	In PERT analysis, when all conditions are least favourable, the time estimate used is: a. Most likely time b. Optimistic time c. Expected time d. Pessimistic time		1

5	A company has a fixed cost of ₹15,000 per month. The variable cost per unit is ₹30, and the selling price per unit is ₹60. What is the minimum production required per month to reach break-even?					1
	a. 400 units	b. 500 units	c. 600 units	d. 700 units		
6	Which ISO standard is certifiable and focuses on quality management systems?					1
	a. ISO 9000	b. ISO 14000	c. ISO 9001	d. ISO 9004		
7	The Fulkerson's Rule is primarily used for:					1
	a. Estimating cost	b. Labelling events in network diagrams	c. Updating resources	d. Project crashing		
8	The difference between total float and free float is known as:					1
	a. Slack	b. Independent Float	c. Lag time	d. Reserve time		
9	The Central Limit Theorem in project management helps in:					1
	a. Determining mean costs	b. Predicting resource allocation	c. Estimating probability of project completion	d. Reducing project scope		
10	Which economic method compares alternatives based on least annual cost?					1
	a. Present Worth	b. Future Worth	c. EUAC (Equivalent Uniform Annual Cost)	d. Internal Rate of Return		
11	A control chart where the process remains within the specified range is said to be:					1
	a. Biased	b. In control	c. Inadequate	d. Rejected		
12	Total Quality Management (TQM) promotes all except:					1
	a. Continuous improvement	b. Customer satisfaction	c. Cost overrun	d. Employee involvement		
Q. 2	Solve the following.					12

A)	Explain Merits and Demerits of Gantt or Bar Chart.		6																				
B)	Differentiate between CPM and PERT.		6																				
Q.3	Solve the following.		12																				
A)	Calculate EST, EFT, LST, LFT, Independent float and interfering float for the activities. Tabulate the result with sample calculation. <table><tr><td>Activity</td><td>1-2</td><td>1-3</td><td>2-4</td><td>3-4</td><td>4-5</td></tr><tr><td>Duration (days)</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr></table>	Activity	1-2	1-3	2-4	3-4	4-5	Duration (days)	2	3	4	5	6		6								
Activity	1-2	1-3	2-4	3-4	4-5																		
Duration (days)	2	3	4	5	6																		
B)	Explain briefly Direct Cost and Indirect Cost for construction project.		6																				
Q. 4	Solve Any Two of the following.		12																				
A)	The time estimates for three activities with a dummy activity is given below. Determine expected time and standard deviation for the path? <table><tr><td>Activity</td><td>to</td><td>tL</td><td>tp</td></tr><tr><td>10-20</td><td>2</td><td>5</td><td>10</td></tr><tr><td>20-30</td><td>0</td><td>0</td><td>0</td></tr><tr><td>30-40</td><td>3</td><td>6</td><td>9</td></tr><tr><td>40-50</td><td>4</td><td>9</td><td>11</td></tr></table>	Activity	to	tL	tp	10-20	2	5	10	20-30	0	0	0	30-40	3	6	9	40-50	4	9	11		6
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10-20	2	5	10																				
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B)	What is the necessity and data required for updating the network?		6																				
C)	Discuss central limit theorem in detail.		6																				
Q.5	Solve Any Two of the following.		12																				
A)	Explain Inflation and Cause of Inflation.		6																				
B)	Explain graphical representation of cost slope.		6																				
C)	A sum of 25000 Rs. becomes 30000 Rs. at the end of 4 years when calculated at simple interest (S.I.). Find out rate (R).		6																				
Q. 6	Solve Any Two of the following.		12																				
A)	Discuss Philosophy of Deming.		6																				
B)	Explain in detail the concept of Quality Control.		6																				
C)	Explain Break Even Analysis with neat Sketch		6																				
	*** End ***																						